Model Documentation

1. Model Used

The model used throughout this project is the Multinomial Naive Bayes (MNB) classifier. It is a probabilistic machine learning model based on Bayes' Theorem and is particularly well-suited for text classification tasks where features represent word counts or TF-IDF scores.

2. Why Multinomial Naive Bayes?

- Efficient for Text Data: Works very well with high-dimensional sparse features produced by TF-IDF vectorization.
- Computationally Fast: Requires less training time compared to models like SVM or Random Forest.
- Good Baseline Model: Provides reliable performance for sentiment and rating prediction tasks.
- Handles Imbalanced Data: Performs reasonably well even when some star ratings are less frequent.

3. Working Principle

Based on Bayes' Theorem:
P(Class|Words) = [P(Words|Class) * P(Class)] / P(Words)

It assumes that the features (words) are independent of each other given the class. For each review, the model calculates the probability of it belonging to each rating class (1★ to 5★) and assigns the class with the highest probability.

4. Why Not Other Models?

- Random Forest / Gradient Boosting: Very powerful for structured/numeric datasets but not efficient for sparse text data.
- Support Vector Machine (SVM): Can achieve high accuracy but computationally expensive for large datasets and requires tuning.

5. Strengths of Naive Bayes in This Project

- Fast to train and predict.
- Works effectively with TF-IDF features.
- Requires minimal parameter tuning.
- Provides interpretable results (class probabilities).

6. Limitations

- Independence assumption (words treated as unrelated) is not always realistic.
- May struggle when features are highly correlated.
- Usually outperformed by deep learning models for very complex tasks.

7. Conclusion

In this project, Multinomial Naive Bayes was chosen because it is simple yet powerful for text-based problems, scalable for large datasets, efficient in computation, and well-suited for review rating prediction tasks. Thus, all model training, evaluation, and inference steps were carried out using Naive Bayes as the core model.